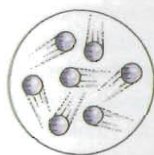


GASES



WHEN YOU CATCH an unpleasant smell given off by a chemistry experiment, your

nose is detecting the presence of a gas released by a chemical reaction. A gas is a type of matter with no fixed shape or volume. Not all gases have a smell, and many are invisible, but all are made up of tiny, fast-moving particles that move rapidly and randomly.



Condensation

Tiny droplets form on the inside of a cold window when water vapour in the air is cooled by the glass and turns into liquid water. The change of a gas to a liquid is called condensation. As a gas cools, its particles lose energy and slow down. The forces between the particles grow stronger and pull them together to form a liquid.

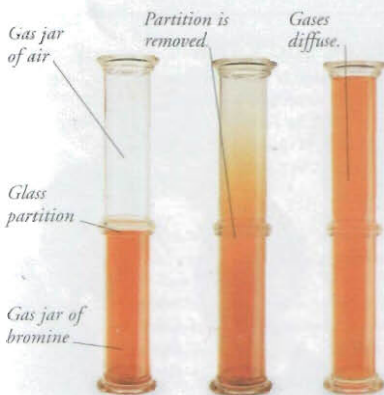
Vapour

A gas normally forms when a liquid boils. However, a type of gas called a vapour sometimes forms when a liquid is below its boiling point. Paint, for example, dries when liquid particles at its surface gain enough energy to escape into the air as a vapour.



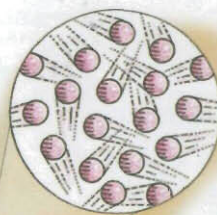
Brownian motion

Dust particles can often be seen dancing in shafts of sunlight. Their random, jittery path is caused by tiny, unseen air molecules that bombard the dust particles. This motion – called Brownian Motion after Robert Brown (1773–1858), a Scottish biologist – shows that gas particles are constantly moving.



Diffusion

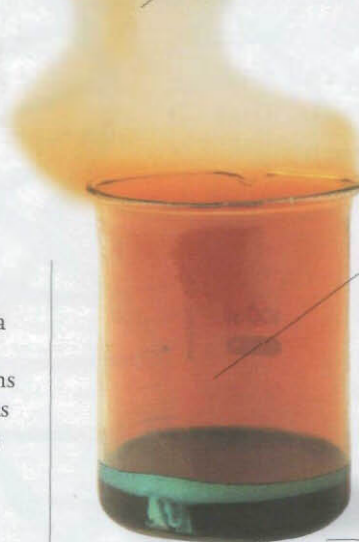
When a jar of bromine gas and a jar of air are placed together, the gases quickly intermingle as their moving particles spread out to fill all the available space. This process is called diffusion. Food cooking in the kitchen is soon smelt throughout the house as gas particles released by the food rapidly diffuse in the air. Diffusion also occurs when solids and liquids dissolve to form solutions.



Gas particles

The forces between the speeding particles of a gas are too weak to hold them in one place, so the gas spreads out.

Nitrogen dioxide gas soon escapes from the beaker and mingles with the air.



Copper and nitric acid react, releasing brown nitrogen dioxide gas.

Properties of gases

A gas quickly spreads out to fill any available space because its free-moving particles travel in all directions. The higher the temperature of a gas, the more energy its particles have and the faster they move. The pressure of a gas is linked to the number of collisions between the gas particles and the walls of its container: the more frequent the collisions are, the greater the pressure the gas exerts.

Amedeo Avogadro

In 1806, an Italian lawyer named Amedeo Avogadro (1776–1856) gave up his legal career to devote himself to the study of physics. In 1811, Avogadro proposed that equal

volumes of all gases at the same temperature and pressure will contain the same number of particles. This is now called Avogadro's Law.



Gas laws

The gas laws are a set of proven theories that allow scientists to predict how a gas will behave when there is a change in its volume, pressure, or temperature. The laws apply only when a gas is held in a sealed container.

The air molecules collide more often with the syringe walls, so the air pressure rises.



Pushing in the plunger reduces the volume.

Charles's Law

This law, formulated by French physicist Jacques Charles (1746–1823), states that the volume of a gas at a constant pressure is proportional to its temperature. Thus, when the temperature of the gas halves, so does its volume.



1 When a gas-filled balloon is cooled in liquid nitrogen at -196°C (-321°F), the gas particles slow down.



2 The particles strike the balloon walls less often, so the gas volume shrinks, and the balloon collapses.



3 As the gas warms again in the air, the gas particles speed up, the volume expands, and the balloon re-inflates.

Boyle's Law

Put your finger over the end of a syringe, push in the plunger, and you will feel the air pressure in the syringe rising. The air obeys Boyle's Law, formulated by the Irish physicist Robert Boyle (1627–91). The law states that when a gas is at a constant temperature, its pressure is inversely proportional to its volume. In other words, if the volume halves, the pressure doubles.



Pressure inside the can blows off the lid.

Pressure Law

Heating a sealed can raises the air pressure inside the can until it is so great that the lid blows off. The air obeys the Pressure Law, which states that when a gas's volume is constant, its pressure is proportional to its temperature. This means that if the gas's temperature doubles, so will its pressure.



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